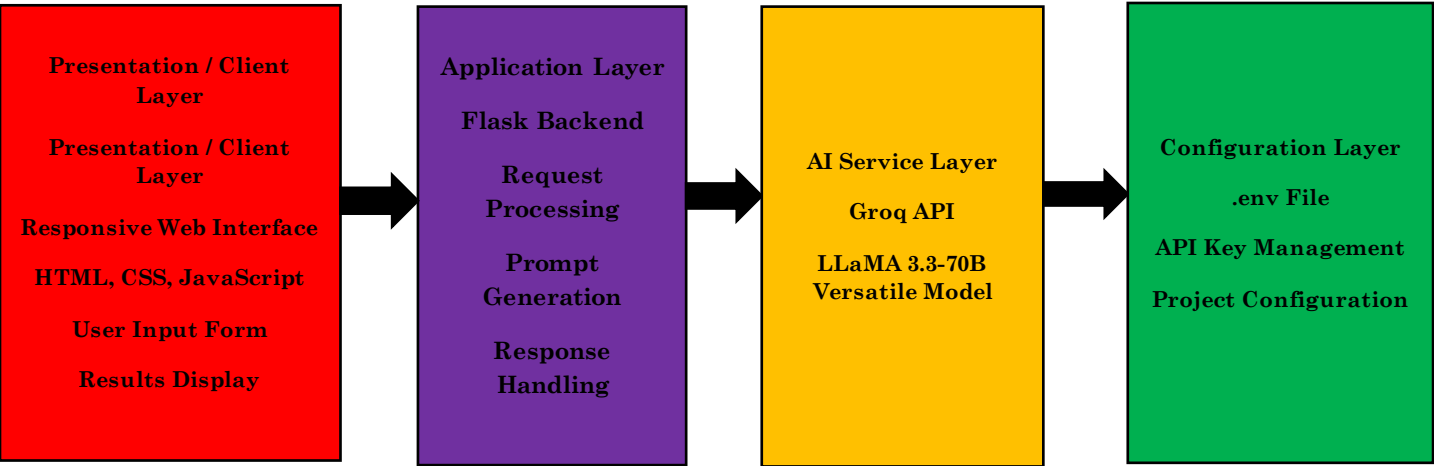


Solution Architecture

Date	July7, 2026
Name	Khushboo Meena
Project Name	CaptionAI: AI-Powered Social Media Caption Generator

Solution Architecture Diagram

The following high-level architecture diagram illustrates the overall structure and workflow of my solution.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface where users enter the campaign description, select the social media platform, choose a writing tone, and view the generated captions, hashtags, and CTA suggestions. It also includes the loading indicator and copy functionality.	HTML5, CSS3, JavaScript

Application Layer	The Flask backend receives user input, validates the request, creates a structured prompt, communicates with the Groq API, processes the AI response, and sends the generated content back to the frontend.	Python, Flask
AI Service Layer	Generates captions, hashtags, and CTA suggestions based on the user's campaign description, selected platform, and writing tone.	Groq API, LLaMA 3.3-70B Versatile
Configuration Layer	Stores the Groq API key securely using the .env file and manages project dependencies and configuration files.	.env, python-dotenv, .gitignore, requirements.txt